

Drug Summary

Selpercatinib (Retevmo) is a highly selective small-molecule inhibitor of the RET (rearranged during transfection) receptor tyrosine kinase ¹. It is FDA-approved for RET-altered cancers (NSCLC, thyroid, and other solid tumors) on the basis of high response rates ². In breast cancer, RET fusions or overexpression are rare (~1.2% of cases ³) but occur across subtypes ⁴. Preclinical and clinical reports show that RET-driven breast cancers (e.g. ER⁺ endocrine-resistant or triple-negative tumors with RET fusions) can be sensitive to RET inhibition ⁵ ⁶. In one case of metastatic RET fusion-positive breast cancer, selpercatinib produced a complete and durable response ⁷. By inhibiting constitutively active RET, selpercatinib blocks downstream signaling (MAPK/ERK and PI3K/AKT) that drives proliferation and survival ¹. Its on-target activity suppresses RET-mediated endocrine resistance in HR⁺ tumors (enhancing estrogen therapy efficacy ⁵ ⁶) and inhibits RET fusion oncogenicity in TNBC/ERBB2⁺ tumors (reducing MAPK/PI3K signaling).

Identifiers & Synonyms

- **ChEMBL ID:** ChEMBL4559134 (Selpercatinib) ⁸.
- **DrugBank ID:** DB15685 ⁹.
- **Synonyms / Brand names:** Selpercatinib (INN); LOXO-292; LY3527723 ¹⁰; Retevmo® (FDA brand); Retsevmo® (EMA brand). Other codes: LY-3527723, LOXO292, WHO10967 ¹⁰ ⁹.

Mechanism of Action (Breast Cancer Context)

Selpercatinib binds directly to the ATP site of RET kinase, preventing its autophosphorylation and activation ¹. In RET fusion-positive cells, this shuts off constitutive RET signaling (which normally drives RAS/MAPK and PI3K/AKT pathways) and leads to reduced cell proliferation and survival ¹. In breast cancer, aberrant RET activity is linked to therapeutic resistance: for example, elevated RET expression in ER⁺ tumors causes resistance to tamoxifen/aromatase inhibitors ⁵. By inhibiting RET, selpercatinib restores sensitivity to endocrine therapy ¹¹. Preclinically, combining selpercatinib with a CDK4/6 inhibitor re-sensitized ER⁺ breast cancer models that had acquired RET-mediated resistance ⁶. Thus, in luminal (HR⁺) breast cancers, selpercatinib's MOA is to abrogate RET-driven bypass signaling (e.g. blocking RET→ERK/PI3K), reversing resistance. In triple-negative or HER2⁺ breast cancers with RET fusions or amplifications, selpercatinib blocks the oncogenic RET kinase itself, collapsing downstream growth pathways ¹ ³.

Primary Targets (Human)

- **RET (HGNC: RET)** – a receptor tyrosine kinase oncogene (“proto-oncogene c-ret”) targeted by selpercatinib ¹². Aberrant RET activity (fusions or amplifications) drives oncogenic MAPK/PI3K signaling in some breast cancers ⁵ ¹.
- **FLT1 (HGNC: FLT1)** – VEGF receptor 1; selpercatinib inhibits VEGFR1 at clinically relevant doses ¹³. (Off-target)
- **FLT4 (HGNC: FLT4)** – VEGF receptor 3; inhibited by selpercatinib ¹⁴. (Off-target)

- **FGFR1 (HGNC: FGFR1)** – Fibroblast growth factor receptor 1; inhibited by selpercatinib ¹⁵ . (Off-target)
 - **FGFR2 (HGNC: FGFR2)** – FGFR2; inhibited ¹⁵ . (Off-target)
 - **FGFR3 (HGNC: FGFR3)** – FGFR3; inhibited ¹⁵ . (Off-target)
- (Note: VEGFR/FGFR inhibition is not the primary clinical effect but may contribute to anti-angiogenesis or toxicity.)

Pathways (Overview)

Key pathways implicated in Selpercatinib's action in breast cancer include those downstream of RET or related to drug response:

- **HALLMARK_KRAS_SIGNALING_UP** – *Receptor-Tyrosine-Kinase/MAPK pathway*. Activated by oncogenic RET, driving proliferation. Selpercatinib *inhibits* this signaling by blocking RET ¹ .
- **HALLMARK_PI3K_AKT_MTOR_SIGNALING** – *PI3K-Akt-mTOR pathway*. Also downstream of RET; promotes survival and growth. RET inhibition by selpercatinib *suppresses* this pathway ¹ . (By contrast, AKT/mTOR activation has been linked to selpercatinib resistance in RET-driven cancers ¹⁶ .)
- **HALLMARK_ESTROGEN_RESPONSE_EARLY** – *Estrogen receptor signaling*. In HR⁺ breast cancer, RET activation can blunt estrogen dependency (promoting endocrine resistance). Selpercatinib *enhances* estrogen therapy efficacy by preventing RET-mediated cross-talk ¹¹ .
- **HALLMARK_EPITHELIAL_MESENCHYMAL_TRANSITION (EMT)** – *EMT/metastasis network*. RET signaling can promote invasion/metastasis; its inhibition may reverse EMT features. (Pathway regulation in Selpercatinib-treated tumors is not well-defined.)
- **REACTOME_SIGNALING_BY_FGFR (R-HSA-190236)** – *Fibroblast growth factor receptor signaling*. FGFR1-3 are weak off-targets of selpercatinib ¹⁵ . Inhibiting FGFR signaling may contribute to anti-tumor or toxic effects.
- **REACTOME_SIGNALING_BY_VEGF (R-HSA-194138)** – *Vascular endothelial growth factor signaling*. VEGFR1/3 are also off-targets ¹⁷ . This may impact angiogenesis.
- **HALLMARK_APOPTOSIS** – *Apoptotic signaling*. Effective RET inhibition often induces tumor cell apoptosis. RET-driven tumors tend to have downregulated apoptosis; selpercatinib is expected to *upregulate* apoptosis. (Evidence in breast cancer context is limited.)
- **HALLMARK_E2F_TARGETS** – *Cell-cycle (E2F) pathway*. RET inhibition has been shown to suppress E2F-driven cell cycle genes in ER⁺ cells ⁶ , reflecting decreased proliferation.
- **HALLMARK_G2M_CHECKPOINT** – *G2/M cell cycle control*. Similarly, selpercatinib affects G2-M regulators in resistant ER⁺ models ⁶ , reducing mitotic progression.
- **HALLMARK_P53_PATHWAY** – *p53 tumor suppressor network*. TP53 mutations frequently co-occur with RET alterations in breast cancer (~80%) ¹⁸ , often attenuating p53-driven apoptosis. This pathway is typically *downregulated* (inactive) in RET-altered tumors, which can contribute to resistance (selpercatinib does not directly reactivate p53).

Detailed Sections

Upregulated pathways (by context/subtype): In RET-driven breast cancers, constitutive RET activity upregulates growth/survival pathways (e.g. MAPK/ERK, PI3K/AKT) ¹ . For example, RET fusion tumors show high KRAS/MAPK signaling. However, specific subtype data (e.g. in TNBC vs. luminal) are limited. No breast-

cancer-specific trials have profiled pathway activation post-selpercatinib, so subtype-specific “upregulated” pathways remain poorly defined.

Downregulated pathways (by context/subtype): Selpercatinib treatment downregulates RET-driven pathways. In ER⁺ tumors, blocking RET may *downregulate* compensatory estrogen-response attenuation (i.e. restore estrogen signaling). In RET fusion cases, selpercatinib *downregulates* MAPK and PI3K/Akt signaling ¹. Apoptosis pathways are expected to be restored (upregulated) after RET inhibition, implying they were downregulated in resistant cells. Again, explicit breast-cancer profiling data are scarce, so this section is inferred from mechanism.

Sensitivity pathways (by context/subtype): Sensitivity pathways are those whose perturbation increases drug efficacy. In ER⁺ breast cancer, intact estrogen receptor signaling can enhance selpercatinib response; combining selpercatinib with endocrine agents (e.g. letrozole) showed synergistic tumor inhibition ¹¹. Thus, an *active estrogen-response pathway* (HALLMARK_ESTROGEN_RESPONSE_EARLY) may sensitize cells to RET inhibition when combined with hormone therapy. Conversely, intact cell-cycle checkpoints (E2F/G2M) in CDK4/6 inhibitor-resistant ER⁺ models were identified as targets of RET inhibition ⁶. Sensitivity pathways for upregulation are not explicitly documented; for downregulation, suppression of PI3K/AKT (e.g. via mTOR inhibitor) was shown to overcome selpercatinib resistance ¹⁶, indicating that downregulating HALLMARK_PI3K_AKT_MTOR enhances selpercatinib sensitivity.

Resistance pathways (by context/subtype): In breast cancer, upregulation of parallel survival pathways can confer resistance. For example, upregulated PI3K/AKT/mTOR signaling (HALLMARK_PI3K_AKT_MTOR) has been linked to selpercatinib resistance ¹⁶. Similarly, activation of growth factor receptors (e.g. MET, EGFR) or cell-cycle drivers can mediate resistance (seen in RET-positive lung/thyroid models). No breast-specific data are reported, but RET-altered breast tumors frequently harbor TP53 co-mutations ¹⁸, potentially enabling evasion of drug-induced apoptosis. Downregulation of tumor suppressor pathways (e.g. p53, apoptosis) can also drive resistance. Thus, upregulation of PI3K/AKT (and possibly compensatory RTK signals) is a key resistance mechanism; downregulation of apoptotic or p53 pathways likely contributes to reduced efficacy. (Evidence in breast is inferred from non-breast RET models.)

Sensitivity pathways for upregulation: No specific pathway has been shown where its *upregulation* in breast cancer directly increases sensitivity to selpercatinib, beyond the general synergy with active ER signaling mentioned above ¹¹.

Sensitivity pathways for downregulation: As noted, downregulating PI3K/AKT/mTOR (e.g. with an mTOR inhibitor) has been shown to re-sensitize RET-driven cells to selpercatinib ¹⁶. Thus, HALLMARK_PI3K_AKT_MTOR_SIGNALING is a candidate – its suppression may sensitize tumors. Inhibiting other parallel pathways (e.g. MAPK) could likewise improve sensitivity, though direct breast data are lacking.

Resistant pathways for upregulation: Upregulation of PI3K/AKT/mTOR is linked to resistance ¹⁶, so HALLMARK_PI3K_AKT_MTOR_SIGNALING is a “resistance pathway” when activated. Other pathways that, if upregulated, could mediate resistance include cell-cycle and anti-apoptosis networks (e.g. HALLMARK_G2M_CHECKPOINT if unchecked cell division persists). Specific context markers in breast cancer (e.g. TP53 loss) support resistance via downregulating HALLMARK_P53_PATHWAY ¹⁸.

Resistant pathways for downregulation: Downregulation of tumor suppressor functions (e.g. HALLMARK_APOPTOSIS, HALLMARK_P53_PATHWAY) is inherently associated with resistance. Breast tumors

with RET alterations often have inactive p53 (TP53 mutations ¹⁸), effectively downregulating this pathway and reducing drug-induced cell death. Loss of pro-apoptotic signaling (not directly documented) would similarly confer resistance to RET inhibition.

Breast Cancer Subtype–Stratified Evidence

- **HR⁺/HER2⁻ (Luminal A/B):** RET is more commonly overexpressed in ER⁺ tumors that have acquired endocrine resistance ⁵ ⁶. High RET expression in metastatic ER⁺ breast cancer correlates with poor response to combined estrogen/CDK4/6 therapy ⁶. Selpercatinib shows activity in this context: e.g. ER⁺ endocrine-resistant models and organoids were resensitized by selpercatinib ⁶. FDA labeling does not yet list breast as an indication, but NCCN guidelines now recognize RET-fusions as actionable in HR⁺ metastatic disease (RET fusion → selpercatinib) ¹⁹.
- **HER2⁺:** RET genomic alterations have been identified in a subset of HER2⁺ tumors ⁵, though clinical data are limited. One index case with HER2⁺ breast cancer had a NCOA4-RET fusion and responded to RET inhibition (cabozantinib) ²⁰ ²¹. The role of selpercatinib specifically in HER2⁺ cases is not documented, but RET-altered HER2⁺ tumors may similarly benefit.
- **Triple-negative (Basal-like/HER2-enriched):** A majority (~65%) of RET-altered breast cancers were ER⁻ ⁴, implying many fall into the triple-negative category. RET fusions or amplifications can drive TNBC growth. Although data in TNBC are sparse, selpercatinib's tumor-agnostic approval suggests any RET fusion TNBC could respond ⁷. To date, no TNBC-specific trial results have been published.
- **BRCA1/2-mutated / HRD⁺:** There is no known interaction between selpercatinib and BRCA status. RET alterations do not specifically associate with BRCA mutations, and no evidence links BRCA/HRD status to selpercatinib response.

Contraindications and Safety (FDA/EMA/PMDA)

- **Contraindications:** None listed for selpercatinib ²².
- **Warnings:** Boxed warnings include *hepatotoxicity* (monitor ALT/AST; dose adjust as needed) and *pneumonitis/ILD* (monitor symptoms; stop drug if severe) ²³. Other warnings: *hypertension* (control BP before/during therapy) ²⁴; *QT prolongation* (avoid in patients with long QT or concomitant QT-prolonging drugs) ²⁵; *hemorrhage* (withhold or discontinue if major bleeding) ²⁶; *impaired wound healing* (stop drug around surgeries) ²⁷; *hypothyroidism* (monitor thyroid function) ²⁸; and *embryo-fetal toxicity* (can cause fetal harm; require contraception) ²⁹ ³⁰.
- **Adverse reactions:** Common AEs (≥25%) include edema, diarrhea, fatigue, dry mouth, hypertension, abdominal pain, constipation, rash, nausea, and headache ³¹. Notable lab abnormalities: elevated ALT/AST, decreased lymphocytes, etc ³². SEL is metabolized by CYP3A4 ³⁰ and is a P-gp/BCRP substrate ³⁰; strong CYP3A modulators can alter its levels.

Trial and Guideline Context

Selpercatinib's efficacy in RET fusion cancers was demonstrated in the phase I/II LIBRETTO-001 trial, which showed an overall response rate of 44% in various RET fusion-positive solid tumors (including breast) ⁷. In LIBRETTO-001, a subset of breast cancer patients with RET fusions achieved clinical responses (2 partial responses reported ³³). There are no breast-specific registration trials, but selpercatinib is recommended in NCCN (breast) guidelines for RET fusion–positive metastatic disease (“useful in certain circumstances”) by

extrapolation ³⁴ . ASCO Post (2025) notes that selpercatinib is guideline-endorsed for RET fusion-positive tumors across tissues ³⁴ . At present, no major breast cancer guideline (NCCN/ESMO/ASCO/NICE) has a separate section for selpercatinib in breast, but molecular testing for RET fusions is implied in advanced HR+ disease due to the NCCN pharmacy updates ³⁵ ³⁴ .

Additional Notes

Selpercatinib is more selective than older multi-kinase RET inhibitors, resulting in better tolerability ³⁶ . It achieves steady-state by ~7 days (half-life ~32 hours) ³⁷ . The drug penetrates the blood-brain barrier, showing activity in CNS RET metastases in other cancers (likely applicable to breast metastases as well). **Mechanistically**, selpercatinib may modestly inhibit other kinases (FGFR1–3, VEGFR1/3) at clinical exposures ¹³ ; the clinical relevance of these off-targets in breast cancer is unclear. In combination studies, adding PI3K/AKT/mTOR inhibitors to selpercatinib is rational to prevent resistance ¹⁶ . No pharmacogenomic biomarkers (besides RET fusion/mutation status) are established for selpercatinib.

Final Pathway Evidence Table

Pathway ID	Regulation (RET context)	Drug Effect	Rationale	Citations
HALLMARK_KRAS_SIGNALING_UP	Up	Inhibited by drug	RET fusions drive RAS/MAPK signaling; selpercatinib blocks RET → downregulates MAPK.	¹
HALLMARK_PI3K_AKT_MTOR_SIGNALING	Up	Inhibited by drug	RET activation triggers PI3K/Akt; selpercatinib suppresses PI3K/AKT pathway.	¹
REACTOME_SIGNALING_BY_FGFR (R-HSA-190236)	Up	Inhibited by drug	FGFR1–3 are inhibited off-target; selpercatinib can block FGFR signaling in tumors.	¹⁵

Pathway ID	Regulation (RET context)	Drug Effect	Rationale	Citations
REACTOME_SIGNALING_BY_VEGF (R-HSA-194138)	Up	Inhibited by drug	VEGFR1/3 are inhibited off-target; selpercatinib may impact VEGF-driven angiogenesis.	17
HALLMARK_E2F_TARGETS	Up	Inhibited by drug	RET inhibition reduces cell-cycle/E2F gene expression in endocrine-resistant ER ⁺ models.	6
HALLMARK_G2M_CHECKPOINT	Up	Inhibited by drug	Selpercatinib suppresses G2/M checkpoint regulators in resistant ER ⁺ cell lines.	6
HALLMARK_ESTROGEN_RESPONSE_EARLY	Down	Sensitivity ↑	Active ER signaling improves endocrine therapy; selpercatinib + letrozole showed synergy.	11
HALLMARK_P53_PATHWAY	Down	Resistance ↑	TP53 is often mutated in RET-altered BC (~80%) ¹⁸ , inactivating p53 pathway.	18

Each entry lists the pathway, how it is dysregulated in RET-driven breast cancer, and how selpercatinib affects it. Citations are provided for all mechanistic claims.

1 8 9 10 12 13 14 15 17 36 37 Selpercatinib: Uses, Interactions, Mechanism of Action | DrugBank Online

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2 7 33 FDA approves selpercatinib for locally advanced or metastatic RET fusion-positive solid tumors | FDA

<https://www.fda.gov/drugs/resources-information-approved-drugs/fda-approves-selpercatinib-locally-advanced-or-metastatic-ret-fusion-positive-solid-tumors>

3 4 5 11 18 20 21 RET rearrangements are actionable alterations in breast cancer | Nature Communications

https://www.nature.com/articles/s41467-018-07341-4?error=cookies_not_supported&code=43cfec8a-b557-4afd-9895-8e55587b34d4

6 19 Frontiers | RET inhibition overcomes resistance to combined CDK4/6 inhibitor and endocrine therapy in ER+ breast cancer

<https://www.frontiersin.org/journals/oncology/articles/10.3389/fonc.2024.1497093/full>

16 Activation of the AKT-mTOR pathway confers selpercatinib resistance in thyroid cancer cells harboring the CCDC6-RET fusion gene - PubMed

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<https://ascopost.com/issues/may-25-2025/nccn-clinical-practice-guidelines-in-oncology-2025-updates/>

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